ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN

QL5 Regular Force Boatswain 00105, NCIOP 00114, Meteorological Technician 00100, Clearance Diver 00342, QL2 Reserve Force Boatswain, NCIOP, Naval Communicator

Navigator's Yeoman Course

AGXO/New Crse ID TBA

Version [01]: Sub-Version [03]
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RECORD OF AMENDMENTS

The "Authority" reference indicates approval of the corresponding chapters of this QSP document. Authority for all Chapters of this QSP resides with the Commanding Officer VENTURE, The Naval Officers Training Centre (VENTURE NOTC), who is both the Training Authority (TA) and the Curriculum Control Authority (CCA). The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J "Course History".

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FOREWORD

1. This training document is developed under the authority of the Commander Royal Canadian Navy and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055-003/PQ-001 Occupation Specialty Specifications (Navigator's Yeoman).

2. This publication is effective on receipt. Performance Objectives (POs) are explained fully in Chapter 2. The Lesson Specifications (LSs)/Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority(ies). QSP revision procedures are outlined in the NTI. Final approval for amendments is the Commanding Officer VENTURE NOTC.

4. Description of Need. HMC Ships require a Navigator's Yeoman to help maintain the navigation outfit and to act as the Navigation Section Storesman. A Navigator's Yeoman is needed to control navigation and hydrographic supplies, maintain chart folios and navigation publications, record information in Officer of the Watch (OOW) notebook, download SHINNADS Dual-M (SDM) voyage log data and perform periodic maintenance on navigation equipment.

5. Scope. The major areas covered by this course are:

   a. correct charts and publications;

   b. control inventory of charts, publications and related materials;

   c. maintain navigation section stores;

   d. maintain navigation equipment;

   e. record information in the OOW notebook; and

   f. set up and secure the bridge.

6. Method of Achieving Objectives. The objectives of the course are met by a combination of theoretical training, practical training and assessment in the classroom and onboard a ship alongside.

7. This QSP shall be used by VENTURE NOTC as the primary authority governing all aspects of the Navigator's Yeoman course.

8. This QSP must be used in conjunction with the Occupation Specialty Specifications in A-PD-055-003/PQ-001. It will also be used in support of validation.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF VENTURE NOTC

AIM

1. HMC Ships require a Navigator's Yeoman to help maintain the navigation outfit. The member must be capable of correcting and updating all navigation charts and publications carried in the ship, and record information in the Office of the Watch notebook and download SDM log data. Additionally, the member must be able to carry out the duties of the Navigation Section's Storesman through controlling navigation and hydrographic supplies, and performing maintenance on navigation equipment. The Navigator's Yeoman Course trains personnel to independently perform the duties of Navigator's Yeoman.

COURSE DURATION CALCULATION

2. Course duration is calculated in accordance with MARCORD 9-30.

Simulator Days = 0.29 days	0.29 days
Class Training Days	8.57 days
Actual Training Days (ATT) p.s. Simulator + Class	8.86 days
Training Administrative Time (TAT) (if required)	0.43 days
Base Administrative Time	0.71 day
Training Travel Time (if required)	0
Course Duration (total calendar days)	12 days

RESOURCES

3. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H. (add extra lines to table as necessary)

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Instructor	LCdr	MARS (MOS-ID 207), DDNO / SPECN (RN)	1	N/A
Instructor	Lt(N)	MARS (MOS-ID 207), FNO	1	N/A
Supply Technician	LS/Cpl	MOS-ID 168	1	N/A
Navigator's Yeoman	LS/Cpl	Navigator's Yeoman qualification	1	Minimum of 1 year experience.

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
HSO Representative	Civilian	ANY	1	Person who works on chart corrections.

4. Identify critical training resources required to conduct the training

CRITICAL TRAINING RESOURCES	NO. OF DAYS

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

   a. Maximum – 10;

   b. Minimum – 6; and

   c. Optimum – 4.

COURSE PREREQUISITES

6. The course prerequisites are as follows:

   a. QL5 Regular Force Boatswain 00105, NCIOP 00114, Meteorological Technician 00100, Clearance Diver 00342 or QL2 Reserve Force Boatswain, NCIOP or Naval Communicator; and

   b. candidate must possess OSS AGJL (Environmental Qualification–Sea/Surface).

RELATED DOCUMENTS

7. Not applicable.

EVALUATION

8. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques. 8. Detailed information about individual evaluation is found in Chapter 3 (Assessment of the Trainee).

VALIDATION

9. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

10. Successful completion of this course constitutes the primary means of attaining the Navigator's Yeoman Crse 108495 (Competency Code AGXO).

PRIOR LEARNING ASSESSMENT

11. Prior Learning Assessment shall be conducted in accordance with A-P9-050-000/PT-012 Canadian Forces Military Equivalencies Program and MARCORD 9-47.

SPECIAL INSTRUCTIONS

12. Pre-course planning and preparation:

TIMESCALE	ACTION / REMARKS
D-12 months	Course dates promulgated.
D-6 months	Produce schedule. Book Navigation and Bridge Simulator (NABS) and Navigational Part Task Trainer (NPTT). Make arrangements with ship(s) for alongside training. Ensure course solicitation message has been sent by NDHQ and NAVRESHQ.
D-3 months	Receive course loading message from NDHQ and NAVRESHQ. Book tour with HSO. Arrange guest lecturer for CF supply lecture.
D-2 months	Send joining letters and instructions. Send introductory letter to ship outlining course aim.
D-4 weeks	Confirm availability of references, stores, classroom and classroom equipment.
D-3 weeks	Book CO, XO, Cox'n, CStdsO and StdsO Nav for welcoming address. Confirm ship(s) for alongside training.
D-1 week	Produce name boards for trainees.

	Prepare classroom, publications and stores required.
D-3 days	Produce class schedule for the classroom and VENTURE's internal website.
D Day	Trainees complete in-routine. Welcoming address.

13. Planning and Programming:

    a. period length – 50 minutes; and

    b. schedule based on 35 periods per week.

14. Course Critiques:

    a. course critiques shall be completed by the course members. Critiques shall be prepared in accordance with VENTURE NOTC's critique SOPs and submitted to the Standards Department at the end of the first week (initial critique), after each phase, and at the end of the course; and

    b. critiques are to be reviewed by all relevant personnel involved with the course.

15. Post-Course Action:

    a. letters of appreciation must be prepared for the Commanding Officer's signature for all guest lecturers and outside agencies that provided tours;

    b. course reports (CF 377) are to be completed by the last day of course;

    c. military achievement certificates are to be presented to all successful trainees; and

    d. complete the following end of course administration:

        (1) Training Statistical Report (lead CTO);

        (2) Excel EC/PC Results spreadsheet (lead CTO); and

        (3) End Course Report and Course Action Plan (SNI and Stds O Nav).

16. Training Support:

    a. three (3) periods are required to conduct a familiarization tour in Hydrographic Service Office (HSO) where candidates are shown the services HSO offers; and

    b. five (5) ITFPR periods are required onboard a ship alongside to demonstrate and test the trainees on how to set up and secure the bridge (EO 003.03 – Set up and secure the bridge of HMC Ships with navigation required material).

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHOTIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF VENTURE NOTC

PO 001 Correct Charts and Publications

1. Performance: Correct Charts and Publications.

2. Conditions:

   a. Given:

      (1) a ship's navigation outfit;
      (2) computer with approved ECPINS software;
      (3) chart correction CDs;
      (4) navigation notices:
         (a) Canadian; and
         (b) British Admiralty.
      (5) navigation warning messages; and
      (6) chart correction equipment and material.

   b. Denied: assistance; and

   c. Environmental: in all Conditions:

3. Standard: The member shall be able to make corrections in accordance with references, to include:

   a. logging all notices in HSO item 565 and 566;

   b. identifying temporary, preliminary and permanent corrections for:

      (1) Canadian charts;

      (2) Canadian publications;

      (3) British Admiralty charts;

      (4) British Admiralty publications;

      (5) American charts; and

      (6) American publications.

   c. correcting charts, to include:

      (1) logging permanent correction numbers in Maritime Command Chart Register (MCCR);

      (2) applying permanent corrections on chart using a fine ballpoint pen;

      (3) logging permanent correction number on bottom left hand corner of chart using a fine ballpoint pen;

      (4) logging temporary or preliminary corrections in HSO item 566 using HB pencil;

(5) applying temporary or preliminary corrections on chart using HB pencil;

(6) logging temporary or preliminary corrections on bottom left hand corner of chart beneath listing of permanent corrections using HB pencils;

(7) accurately locating the position of each correction on the chart using:

(a) range and bearing;

(b) latitude and longitude; and

(c) template.

(8) correcting charts accurately using:

(a) fine ballpoint pen;

(b) block corrections;

(c) HB pencil for temporary and preliminary corrections; and

(d) electronic methods.

d. correcting publications, to include:

(1) making corrections by:

(a) using a fine ball point pen; and

(b) photocopying and pasting the original text correction.

(2) logging corrections using a fine ball point pen in:

(a) the cover of the publication after the corrections are complete; and

(b) HSO item 566 after the corrections are complete.

e. working with navigation warning messages to include:

(1) determining the purpose of the message;

(2) logging and sorting messages;

(3) filing messages in numerical order; and

(4) applying corrections.

4. <u>References.</u>

A1, A4, , , A5, A6, A7, A8, , , A11.

5. <u>Specification Task Numbers.</u>

1, 3.

6. Supporting Skills.

1, 2, 3, 4, 6.

7. Supporting Knowledge.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14.

8. Training Limitations.

Nil.

PO 002 Control and Maintain Navigation Material.

1. Performance: Control and Maintain Navigation Material.

2. Conditions:

   a. given:

      (1) a ship's navigation outfit;

      (2) navigation Supply Customer Account (SCA) ;

      (3) direction from the Navigating Officer regarding type/quantity of stores required;

      (4) supply forms 2302;

      (5) identity and condition tag CF 942;

      (6) temporary loan cards (DND 638);

      (7) equipment on the Navigation Section Supply Customer Account (SCA);

      (8) cleaning material;

      (9) lubricating material;

      (10) appropriate tools;

      (11) ship's logistics support; and

      (12) services of HSO.

   b. denied: assistance; and

   c. environmental: in all conditions.

3. Standard: The member shall be able to:

   a. maintain control of navigation related material in accordance with references by:

      (1) mustering the navigation outfit;

      (2) assessing quality / usability of charts and publications;

      (3) using DND e-mail:

         (a) for return or exchange of publications that are unserviceable;

         (b) to demand:

            i. charts;

            ii. publications;

iii. air charts;

iv. maps; and

v. HSO stationary.

(4) securing charts, publications and HSO items for sea to prevent loss and damage; and

(5) safely stowing and handling of classified charts.

b. maintain the Navigation Section's stores in accordance with references, to include:

(1) maintaining the stores file / book including a record of the:

(a) date of demand;

(b) item ordered;

(c) stock number;

(d) document number;

(e) date received;

(f) quantity ordered; and

(g) class of item.

(2) mustering of stores and navigation equipment, ensuring:

(a) all items are sighted;

(b) all items are serviceable;

(c) all items are accounted for; and

(d) report of deficiencies is made to the account holder.

(3) demanding stores and equipment, ensuring:

(a) the demand form is accurately and legibly completed;

(b) the information is recorded in the stores file; and

(c) a local purchase order (LPO) is made.

(4) returning equipment, ensuring the:

(a) condition tag is accurately and legibly completed;

(b) demand form is accurately and legibly completed; and

(c) information is recorded in the stores file.

(5) issuing equipment, ensuring:

(a) temporary loan cards are accurately and legibly completed; and

(b) the loan card is filed.

c. maintain navigation equipment in accordance with the references, to include:

(1) storing instruments, ensuring:

(a) they are placed in the correct case;

(b) they are secured for sea; and

(c) the case is secured to prevent loss or damage.

(2) inspecting for:

(a) cleanliness; and

(b) damage.

(3) cleaning:

(a) lens and mirrors clear of obstruction;

(b) to ensure there is no salt build up on the equipment; and

(c) the equipment free of rust corrosion.

(4) lubricating to ensure all parts move freely;

(5) performing periodic maintenance to ensure:

(a) all screws are tightened; and

(b) all straps are secured.

(6) using the CF Supply System to ensure the correct procedure is used for returning items for:

(a) repair; and

(b) replacement.

4. <u>References.</u>

A1 to A10, A12

5. <u>Specification Task Numbers.</u>

2 to 5

6. Supporting Skills.

3, 5, 6

7. Supporting Knowledge.

2,7,11 to 22.

8. Training Limitations.

Nil.

PO 003 Execute the Navigating Officer's Requirements for Navigation Materials & Equipment

1. Performance: Execute the Navigating Officer's Requirements for Navigation Related Materials & Equipment on the Bridge of HMC Ships.

2. Conditions:

   a. Given:

      (1) a ship alongside with a complete navigation outfit;

      (2) a ship conducting a transit;

      (3) OOW notebook;

      (4) Computer with approved ECPINS software;

      (5) CD ROMs;

      (6) information for entries; and

      (7) bridge set up/secure checklist.

   b. denied: assistance; and

   c. environmental: in all conditions.

3. Standard: The member shall be able to:

   a. record information in the OOW notebook in accordance with the references to include:

      (1) formatting the notebook;

      (2) correcting mistakes;

      (3) recording applicable entries as directed by the Officer of the Watch (OOW) or Navigator to include:

         (a) four-figure time;

         (b) courses;

         (c) speeds;

         (d) fixes;

         (e) ship's activities;

         (f) manoeuvring orders;

         (g) weather; and

(h) miscellaneous entries.

(4) conducting a time check.

b. maintain SDM records to include:

(1) downloading the voyage log data onto CD ROMs;

(2) storing the CD ROMs until the ship's logs have been sent to Director History and Heritage (DHH); and

(3) destroying the CD ROMs after the ship's logs have been sent to DHH.

c. set up the bridge of HMC Ships in accordance with ships' checklists, to include:

(1) ensuring all necessary equipment is in place and serviceable prior to sailing; and

(2) ensuring all publication, charts, and stationary required for sailing are on the bridge.

d. secure the bridge of HMC Ships in accordance with ships' checklists, to include:

(1) stowing all equipment, publications and stationary when Special Sea Duty Personnel (SSD) are secured and the vessel is alongside; and

(2) scrubbing all charts and putting them back into their correct folios as directed by the Navigating Officer.

4. References:

A1, A2, A5, A6, A9, A12, A11.

5. Specification Task Numbers:

4, 6, 7.

6. Supporting Skills:

3, 6, 7.

7. Supporting Knowledge:

7, 15, 19, 22 to 30.

8. Training Limitations: Nil.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BY GRANTED BY THE COMMANDING OFFICER OF VENTURE NOTC

GENERAL

1. All Performance Checks (PCs) and Performance Checklists (PCLs) are based on Performance Objectives listed in Chapter 2 of this QSP and must be successfully completed to achieve the qualification. A Pass/Fail assessment shall be made on each candidate. Enabling Checks (ECs) and Enabling Checklists (ECLs) are used to help the trainee work towards meeting the standard required in the PCs/PCLs in each Performance Objective (PO).

PERFORMANCE CHECKS

2. In accordance with MARCORD 9-60 Volume 1, trainees are permitted to attempt one supplementary PC/PCL (re-test) in a maximum of two different POs. Failure of a third PC/PCL will result in a TRB. Failure of a supplementary PC/PCL (re-test) will constitute failure in that PO and will result in the trainee appearing before a TRB. A further attempt at passing a PC/PCL may be granted if, in the judgement of the TRB members, unusual circumstances exist. These circumstances shall be recorded on the course report.

3. The following chart summarizes the approach to PO assessment during the Navigator's Yeoman course:

PO/EO	ASSESSMENT	REMARKS AND REQUIREMENTS
PO 001 – Correct Charts & Publications	PC 001	Written PC; pass mark is 80%.
EO 001.01 - Manually Correct Charts & Publications	EC 001.01	Written EC; pass mark is 80%.
EO 001.02 – Correct Electronic Charts	EC 001.02	Written EC; pass mark is 80%.
PO 002 – Control and Maintain Navigation Material	PC 002	Written PC; pass mark is 70%.
EO 002.01 – Control Charts, Publications & Other Navigation Related Material	Nil	No EC for this EO.
EO 002.02 – Maintain the Navigation Section's Stores	EC 002.02	Written EC; pass mark is 70%.
EO 002.03 – Maintain the Navigation Equipment	EC 002.03	Written EC; pass mark is 70%.
PO 003 – Execute the Navigating Officer's Requirements for Navigation Related Materials & Equipment	PCL 003	Written PC (covering material for EO 003.01 and EO 003.02); pass mark is 70% and PCL 003 Checklist must be rated as Satisfactory in order

		to pass.
EO 003.01 – Maintain the OOW Notebook	Nil	No EC for this EO.
EO 003.02 – Maintain SDM Records	Nil	No EC for this EO.
EO 003.03 – Set up and secure the bridge Navigation Equipment.	Nil	No EC for this EO.

4. The pass marks for the ECs (Enabling Checks) are outlined in the table in paragraph 3. If a trainee fails a supplementary EC (re-test) it is up to the discretion of the Senior Navigation Instructor (SNI) to determine whether the trainee requires additional re-test(s) in order to be successful in the PC for that PO. The ECs are not pre-requisites for the trainee to progress on to subsequent EOs or POs. For example, if a trainee failed EC 001.02 and then failed the re-test, the trainee could still continue with training and challenge PC 001.

5. The pass marks for all academic PCs (Performance Checks) are outlined in the table in paragraph 3. Failure of a supplementary PC or PCL (Performance Check List) will result in appropriate administrative action as outlined in paragraphs 2 and paragraphs 11 through 13 of this chapter. Failure of a supplementary PC/PCL (re-test) constitutes failure of the entire PO.

PROGRESS MONITORING

6. Trainee's progress shall be continually monitored and recorded to provide:

   a. feedback on effectiveness of training;

   b. early warning of trainee difficulties;

   c. equal distribution of training assignments;

   d. if required, information for Training Review Boards (TRBs); and

   e. information to assist in completion of the course reports (CF 377).

7. The instructor shall maintain, in the trainee's training file, the results of:

   a. ECs;

   b. PCs/PCLs;

   c. exercises/assignments;

   d. any significant incident that indicates the trainee has a negative attitude or lacks professionalism; and

   e. all counselling sessions or interviews.

8. In the event of an EC/PC/PCL failure, the trainee will be counselled and given, as appropriate:

a. additional training;

b. additional assignments;

c. additional study; and/or

d. a re-test for the EC/PC/PCL.

UNSATISFACTORY TRAINING PROGRESS

9. The training review process is contained in MARCORD 9-60 Volume 1. 9. Unsatisfactory course performance is indicated by:

a. failure of a PC/PCL;

b. failure of an EC re-test; and

c. development of an unprofessional attitude.

10. Every effort must be made to assist a trainee having difficulty before it becomes necessary to convene a TRB. However, unsatisfactory course progress, as outlined in paragraph 9, may result in a TRB in accordance with guidelines set out in paragraphs 11 through 13.

TRAINING REVIEW BOARD

11. Training Review Board requirements are contained in MARCORD 9-60 Volume 1.

12. A TRB shall be convened if the trainee:

a. fails a supplementary PC/PCL (re-test);

b. develops an unprofessional attitude;

c. is given a medical category that prohibits further training; or

d. requests to cease training.

13. If a TRB is convened, the members of the Board shall review:

a. the trainee's training file;

b. performance on ECs and PCs/PCLs;

c. overall behaviour and attitude during the course;

d. study habits;

e. overall course performance;

f. indication of earlier trouble and action taken to rectify problems; and

g. feasibility and suitability of the trainee to continue with training.

14. Course removal for medical, compassionate or other reasons will require the convening of a TRB as prescribed in MARCORD 9-60.

COURSE REPORTS

15. Course reports (CF 377) are completed IAW CFAO 26-12. 14. A Canadian Forces Course Report (CF 377) shall be prepared and distributed for each trainee in accordance with DAOD 5031-9. Completed course reports shall be distributed as follows:

a. Regular Force members:

(1) original to NDHQ;

(2) second copy to the trainee's parent unit;

(3) third copy to be retained by VENTURE; and

(4) fourth copy to the trainee;

b. Naval Reserve members:

(1) original to Naval Reserve Headquarters;

(2) second copy to the trainee's parent unit;

(3) third copy to be retained by VENTURE; and

(4) fourth copy to the trainee.

16. For trainees who pass a PC re-test, the CF 377 shall reflect the minimum pass mark for this PC, not the mark they achieved on the re-test.

17. Successful candidates are assessed as 'Pass' and unsuccessful candidates are assessed as 'Fail'. Letter Grade and Class Standing are not recorded on the CF 377 for this course.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BY GRANTED BY THE COMMANDING OFFICER OF VENTURE NOTC

PO 001 Correct Charts and Publications.

EO 001.01 Manually correct charts and publications

1. Performance: Manually correct charts and publications.

2. Conditions:

   a. given:

      (1) a ship's hydrographic outfit; and

      (2) chart correction equipment and material.

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to manually apply corrections to charts and publications neatly and accurately without omission by:

   a. determining from navigation notices and navigation warning messages which corrections to apply;

   b. locating the position of corrections on charts;

   c. making permanent corrections to charts using pen or block correction method;

   d. logging and applying temporary and preliminary corrections to charts using pencil;

   e. logging the correction number on the bottom left corner of the chart;

   f. logging corrections properly in MCCR;

   g. correcting publications using pen, pencil or block correction method; and

   h. logging corrections on the inside cover of publications.

4. Time: There are 16 x 50 minute periods, which are broken down as follows:

   a. 2 periods – geodesy;

   b. 1 period – chart symbols;

   c. 2 periods – manual chart corrections;

   d. 3 periods – tour of HSO;

   e. 4 periods – EC;

   f. 1 period – post-EC review;

g. 2 periods – PC; and

h. 1 period – post-PC review.

5. Method/Media:

a. lectures; and

b. practical exercises.

6. References:

A1, A4, , , A5, A6, A7, A8, , A11.

7. Training Aids:

a. ship's navigation outfit including duplicate charts;

b. navigation notices:

(1) Canadian Coast Guard Notice to Mariners;

(2) DND Notice to Mariners; and

(3) British Admiralty Notice to Mariners.

c. navigation warning messages;

d. chart work equipment to include:

(1) parallel ruler;

(2) compass and dividers;

(3) color ink pens (red, green, magenta); and

(4) HB pencils and white erasers.

e. flat smooth work area;

f. classroom with PowerPoint; and

g. whiteboard.

8. Test Details:

a. EC 001.01 named "Manual Chart Correction Exercise", it is written in four 50 minute periods and covers theory, chart symbol identification, and practical correction tasks for manually correcting charts and publications. This EC is an assignment done during class time; and

b. PC 001 assesses material covered in EO 001.01.

9. Remarks:

None.

10. Teaching Points:

a. describe position, direction and distance on the Earth's surface to include:

(1) defining the following:

(a) the Poles;

(b) the equator;

(c) latitude and longitude;

(d) sea mile;

(e) nautical mile;

(f) a cable; and

(g) true and relative bearing.

(2) positions expressed in:

(a) latitude and longitude; and

(b) range and bearing from a known point.

b. identify chart symbology and buoyage to include:

(1) the standard symbols, abbreviations and terms described in:

(a) Canadian chart 1;

(b) US chart 1; and

(c) British Admiralty chart 5011.

(2) the IALA buoyage system to identify:

(a) lateral marks;

(b) cardinal marks;

(c) isolated danger marks;

(d) safe water or fairway marks;

(e) ODAS buoys; and

(f) miscellaneous marks to include:

i. mooring buoys;

ii. day beacons; and

iii. leading beacons.

(3) describe and give examples of the various shapes, colors and light characteristics used for navigation marks;

c. draw chart symbols by hand to include:

(1) locating chart symbol types;

(2) determining the correct size and shape;

(3) using the appropriate color ink; and

(4) neatness and accuracy.

d. corrections will be applied IAW refs to include:

(1) logging all notices in MCCR;

(2) identifying temporary, preliminary and permanent corrections for Canadian, US and British Admiralty charts to include:

(a) properly logging permanent corrections in MCCR;

(b) applying permanent corrections to charts using a fine ballpoint pen;

(c) logging permanent corrections on the bottom left hand corner of charts using a fine ballpoint pen;

(d) logging temporary and preliminary corrections in MCCR using an HB pencil;

(e) applying temporary and preliminary corrections to charts using an HB pencil;

(f) logging temporary and preliminary corrections on the bottom left hand corner of the chart beneath listing of permanent corrections using HB pencil;

(g) accurately locate positions for corrections on the chart using:

i. range and bearing;

ii. latitude and longitude; and

iii. template

(h) correcting charts accurately using:

i. a proper color fine ballpoint pen;

ii. block corrections; and

iii. a HB pencil for temporary and preliminary corrections.

(3) identifying temporary, preliminary and permanent corrections for Canadian, US and British Admiralty publications to include:

(a) correcting publications using:

i. fine ball point pen; and

ii. block corrections.

(b) logging the correction in the appropriate section of the publication; and

(c) once the correction has been applied log the correction in MCCR using a fine ballpoint pen.

e. work with navigation warning messages to include:

(1) determining the purpose of the message to identify:

(a) whether a correction is to be made; or

(b) whether the message is an advisory to shipping.

(2) logging and sorting messages into:

(a) CANHYDROLANT/PAC;

(b) HYDROLANT/PAC;

(c) NAVAREA; and

(d) navigation warning summaries.

(3) filing messages in numerical order;

(4) briefing the Navigating Officer on priority corrections;

(5) applying corrections;

(6) identify and give examples of the type of information contained in navigation warning messages; and

(7) identify how long messages are to be retained.

ENABLING CHECK - EC 001.01

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One EC per trainee.

b. Classroom.

c. Paper charts.

d. Publications

e. Extracts from Canadian Coast Guard Notice to Mariners (CCG NTM).

f. Extracts from British Admiralty Notice to Mariners.

g. Extracts from DND Notice to Mariners.

h. Extracts from Maritime Command Chart Register (MCCR).

i. Pencils.

j. Erasers.

Arrangements And Procedures

1. This EC is an assignment done during class Time: It consists of written questions and a practical exercise to assess the trainee's knowledge of chart and publication corrections and skill on how to do chart and publication corrections

2. Time: 4 periods (3 hour and 20 minutes).

3. Pre-Test Instructions for the CTO:

   a. trainees are to write their answers right on the EC and chart; and

   b. pass mark is 80%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of EC:

   a. knowledge;

   b. skill;

   c. in-class assignment;

   d. closed book except for the practical (skill) portion of the assignment, where references, as listed in this EC under material required, are allowed; and

   e. short answer questions.

2. Answers: All answers are to be written on the EC below the questions and on the chart.

3. Time Limit: 4 periods (3 hour and 20 minutes).

4. Pass Mark: 80%.

PART III - ENABLING CHECK NOTES

The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
Geographical measurement	10%
Chart symbols	10%
Navigation messages	10%
Publications	10%
Corrections (theory):	10%
Corrections (practical/skill)	50%

PART IV – ENABLING CHECK LIST

Not Used.

EO 001.02 Correct Electronic Charts.

1. Performance: Correct Electronic Charts.

2. Conditions:

   a. given:

      (1) a ship's hydrographic outfit; and

      (2) laptop with approved ECPINS software;

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to apply electronic chart corrections without omission by:

   a. determining the appropriate corrections CD to use and then applying the corrections to the chart; and

   b. logging corrections properly in the Maritime Command Chart Register (MCCR).

4. Time: There are 20 x 50 minute periods, which are broken down as follows:

   a. 14 periods – how to operate SDM and implement chart corrections;

   b. 2 periods – Electronic chart correction practical

   c. 3 periods – EC 001.02 In class practical; and

   d. 1 period – post-EC review.

5. Method/Media:

   a. lectures;

   b. Navigational Part Task Trainer (NPTT); and

   c. practical exercises.

6. References:

A1, A4, , , A5, A6, A8, , , A11.

7. Training Aids:

   a. ship's navigation outfit;

   b. navigation notices:

      (1) Canadian Coast Guard Notice to Mariners;

(2) DND Notice to Mariners; and

(3) British Admiralty Notice to Mariners.

c. chart corrections CDs;

d. flat smooth work area;

e. classroom with PowerPoint;

f. whiteboard;

g. ECPINS-M console (full or laptop version);

h. NPTT; and

i. MCCR.

8. Test Details:

a. EC 001.02, named “Electronic Chart Correction Exercise”, is written in six 50 minute periods and covers theory and practical correction tasks relating to electronic. This EC is an assignment done during class time; and

b. PC 001 assesses material covered in EO 001.02.

9. Remarks: None.

10. Teaching Points:

a. definitions and abbreviations to include:

(1) International Maritime Organization (IMO);

(2) International Hydrographic Organization (IHO);

(3) Canadian Hydrographic Service (CHS);

(4) Electronic Chart Display and Information System (ECDIS);

(5) Electronic Chart Precise Integrated Navigation System (ECPINS);

(6) Digital Nautical Chart (DNC);

(7) World Geodetic Survey 84 (WGS 84); and

(8) Global Positioning System (GPS).

b. system components to include:

(1) three main components; and

(2) types of data.

c. electronic charts to include:

(1) who maintains the database; and

(2) who produces individual charts.

d. chart database to include:

(1) raster charts;

(2) vector charts; and

(3) S-57 charts.

e. ECDIS requirements to include:

(1) vector format electronic charts;

(2) digital chart data transfer;

(3) warnings and alarms;

(4) priority layers;

(5) calculation requirements to include:

(a) geographical position to display coordinates;

(b) local datum to WGS 84; and

(c) true distance and azimuth between geographic points.

f. the integrated bridge to include:

(1) computer processor;

(2) chart database;

(3) system display;

(4) planning station;

(5) control system; and

(6) radar.

g. electronic chart corrections to include:

(1) interactive entry to include:

(a) how corrections are entered; and

(b) its advantages/disadvantages.

(2) semi-automatic entry to include:

(a) how corrections are entered; and

(b) its advantages/disadvantages.

(3) automatic entry to include:

(a) how corrections are entered; and

(b) its advantages/disadvantages.

h. current status of electronic chart implementation and its relevance to:

(1) Canadian naval ships;

(2) merchant ships;

(3) distribution of vector and S-57 charts; and

(4) British Admiralty charts available.

i. Integrated Navigation Systems to include:

(1) GPS, DGPS, and Loran C; and

(2) how systems operate.

j. real time water levels;

k. problems with chart data; and

l. legal aspects of passage planning.

ENABLING CHECK – EC 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

a. One EC booklet per trainee.

b. Classroom.

c. ECPINS laptop.

d. Electronic charts.

e. Extracts from Canadian Coast Guard Notice to Mariners (CCG NTM).

f. Extracts from British Admiralty Notice to Mariners.

g. Extracts from DND Notice to Mariners.

h. Extracts from Maritime Command Chart Register (MCCR).

i. Pencils.

j. Erasers.

Arrangements and Procedures

1. This EC is an assignment done during class time. It consists of written questions and a practical exercise to assess the trainee's knowledge of electronic charts and skill on how to do electronic chart corrections.

2. Time allotted: 3 periods (2 hour and 30 minutes).

3. Pre-Test Instructions for the CTO: trainees are to write the short answers right on the EC and make chart corrections using their laptops; and

4. Pass mark is 80%.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. Type of EC:

   a. knowledge;

   b. skill;

   c. in-class assignment;

   d. closed book except for the practical (skill) portion of the assignment, where references, as listed in this EC under material required, are allowed; and

   e. short answer questions.

2. Answers: All answers are to be written on the EC below the questions or saved on the electronic chart.

3. Time Limit: 3 periods (2 hour and 30 minutes).

4. Pass Mark: 80%.

PART III – ENABLING CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
Use of SDM	15%
SDM Policies	15%
Electronic Chart Corrections	70%

PART IV – ENABLING CHECK LIST

Not Used.

PERFORMANCE CHECK - PC 001

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One PC per trainee.

b. Classroom.

c. Electronic charts.

d. Paper charts.

e. Extracts from Canadian Coast Guard Notice to Mariners (CCG NTM).

f. Extracts from British Admiralty Notice to Mariners.

g. Extracts from DND Notice to Mariners.

h. Extracts from Maritime Command Chart Register (MCCR).

i. ECPINS laptop.

j. Pencils.

k. Erasers.

Arrangements And Procedures

1. This written knowledge PC is used to assess the trainee's knowledge in policies and procedures in implementing corrections in paper and electronic charts. It covers material from EO 001.01 and 001.02.

2. Time: 2 periods (1 hour 40 minutes).

3. Pre-Exam Instructions for the CTO:

   a. trainees are to write their answers right on the PC;

   b. references are not allowed; and

   c. pass mark: 80%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of PC:

   a. knowledge;

   b. written;

   c. closed book; and

d. short answer questions.

2. Material Covered: EO 001.1 and 001.2.

3. Answers: All answers are to be written on the PC below the questions.

4. Time Limit: 2 periods (1 hour 40 minutes).

5. Pass Mark: 80%.

PART III - PERFORMANCE CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
Theory – background knowledge on chart corrections	20%
Manual chart corrections	40%
Electronic chart corrections	40%

PART IV – PERFORMANCE CHECK LIST

Not Used.

PO 002 Control and Maintain Navigation Material.

EO 002.01 Control Charts, Publications & Other Navigation Related Material

1. Performance: Control of charts, publications and other navigation related material.

2. Conditions:

   a. given:

      (1) a ship's navigation outfit; and

      (2) services of HSO.

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to maintain control of charts, publications and other navigation related material in accordance with the references, to include:

   a. mustering the navigation outfit;

   b. assessing the quality and usability of charts and publications;

   c. securing navigation stores for sea;

   d. safely stowing and handling of classified charts; and

   e. assist in preparing for navigation administrative inspections.

4. Time: There are 2 x 50 minute periods, which are broken down as follows:

   a. 1 period – responsibilities of the Navigator's Yeoman; and

   b. 1 period - Navigation administration and Navigation inspections.

5. Method/Media: Lectures.

6. References:

A1, A2, A4, A5, A6, A7, , , A10.

7. Training Aids: Classroom with Power Point.

8. Test Details: There is no EC for this EO. The material covered in this EO is assessed in PC 002.

9. Remarks: None.

10. Teaching Points:

a. describe mustering of a navigation outfit to include:

(1) conducting a stores check; and

(2) mustering of the chart outfit and publications.

b. assessing quality and usability of charts and publications to determine:

(1) when charts/publications need to be replaced;

(2) when old charts may be discarded;

(3) how to properly discard unclassified and classified charts; and

(4) who to contact when unsure of chart's usability/quality.

c. describe and demonstrate how to correctly demand hydrographic products to include:

(1) how to distribute demands and exchange requests for:

(a) charts;

(b) publications;

(c) air charts;

(d) maps; and

(e) HSO stationary.

(2) how to correctly track and log all demands and exchange requests.

EO 002.02 Maintain the Navigation Section's stores

1. Performance: Maintain the Navigation Section's stores.

2. Conditions:

   a. given:

      (1) direction from the Navigating Officer regarding type/quantity of stores required; and

      (2) ship's logistics support;

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to maintain the Navigation Section's stores to include:

   a. maintaining the stores file;

   b. completing a muster of the navigation stores;

   c. conducting a Supply Customer Account (SCA) verification muster;

   d. demanding stores and equipment using the required documentation;

   e. returning equipment using the required documentation; and

   f. ensuring the proper issue of equipment within a ship.

4. Time: There are 4 x 50 minute periods, which are broken down as follows:

   a. 2 periods – CF supply system (given by a guest lecturer);

   b. 1 period – EC; and

   c. 1 period – post-EC review.

5. Method/Media:

   a. lectures; and

   b. practical exercises.

6. References:

A1, A3.

7. Training Aids:

   a. navigation section Supply Customer Account (SCA);

b. CF supply form DND 2227;

c. identity and condition tag CF 942;

d. temporary loan cards DND 638; and

e. classroom with PowerPoint.

8. Test Details:

a. EC 002.02 named “Navigation Section’s Stores Exercise”, is written in one 50 minute period and consists of a practical/skill exercise; and

b. PC 002 assesses material covered in EO 002.02.

9. Remarks: None.

10. Teaching Points:

a. describe the Canadian Forces Supply System to include the following definitions:

(1) Supply Customer Account (SCA);

(2) stock code;

(3) requisition;

(4) expense element;

(5) unit district codes;

(6) warehouse codes to include:

(a) main warehouse;

(b) MSA (Material Service Account)

(c) spare parts; and

(d) repair;

(7) material priority codes;

(8) stock holding codes to include:

(a) available for issue (AV);

(b) contingency stock (CS);

(c) repairable (RP); and

(d) restricted (RD).

(9) accountability codes to include:

(a) accountable equipment (A);

(b) consumable (C);

(c) disposable (D);

(d) controlled equipment (E);

(e) obsolete (O); and

(f) services (S).

(10) rotation requisition;

b. describe Supply Customer Account (SCA) holder responsibilities to include:

(1) safeguarding material;

(2) periodic verification of SCA holdings;

(3) immediately reporting losses;

(4) repair or replace unserviceable material;

(5) returning surplus material to MAROPSGROUP 4/5 Supply

(6) maintaining a current SCA report;

(7) maintaining a manual record of transactions; and

(8) maintaining document number control;

c. describe the mustering of a navigation outfit to include:

(1) maintaining a stores file/book that includes the following:

(a) date of demand;

(b) item ordered;

(c) stock number;

(d) document number;

(e) date received;

(f) quantity ordered; and

(g) class of item.

(2) mustering of stores and navigation equipment to ensure:

(a) all items are:

i. sighted;

ii. serviceable; and

iii. accounted for.

(b) all deficiencies are reported to the account holder.

(3) demanding stores and navigation equipment to include:

(a) identifying the proper form to be used;

(b) identifying stock code information;

(c) completing the demand form CF 2227;

(d) recording information in the stores file; and

(e) following the required procedures to make local purchase orders.

(4) returning navigation equipment to include:

(a) identifying when/why items need to be returned;

(b) the process for returning items to include:

i. arranging for pick up/delivery with supply group;

ii. completing return voucher form CF 2227;

iii. completing identification and condition tag form CF 942;

iv. distributing copies of the forms; and

v. ensuring a credit return document is received.

(c) recording information in the Section's stores file;

(5) issuing of equipment on temporary loan to include:

(a) safe stowage of equipment;

(b) using temporary loan cards for issue of equipment;

(c) accurately and legibly completing temporary loan cards; and

(d) filing and tracking of loan cards.

(6) correctly submitting loss reports to include:

(a) conducting a thorough search for the item;

(b) raising the loss report after 7 days;

(c) providing a detailed narrative of the circumstances surrounding the loss; and

(d) submitting the loss report to MAROPSGROUP 4/5 SUPPLY Logistics Officer for action.

ENABLING CHECK - EC 002.02

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One EC booklet per trainee.

b. Classroom.

Arrangements And Procedures

1. This EC consists of written questions and a practical exercise to assess the trainee's knowledge of how to maintain the Navigation Section's stores.

2. References are not allowed.

3. Time allotted: 1 period (50 minutes).

4. Pre-Test Instructions for the CTO:

   a. trainees are to write their answers right on the EC;

   b. references are not allowed; and

   c. pass mark is 70%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of EC:

   a. knowledge;

   b. written;

   c. closed book; and

   d. short answer questions.

2. Time Limit:1 period (50 minutes).

3. Pass Mark: 70%.

PART III - ENABLING CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
General CF Supply System	25%
Practical Applications	75%

EO 002.03 Maintain the navigation equipment

1. Performance: Maintain the navigation equipment.
2. Conditions:
   a. given:
      (1) Stuart's Distance Meter;
      (2) Binoculars;
      (3) Stevenson Screen;
      (4) Night Vision Goggles;
      (5) Barographs;
      (6) Weymouth-Ross Distance Meter;
      (7) Azimuth Circle;
      (8) Sextant (A/B class);
      (9) Three-Arm Station Pointer; and
      (10) access to an ECDIS console with ECPINS-M software installed.
   b. denied: assistance; and
   c. environmental: in all Conditions:
3. Standard: The trainee shall be able to maintain navigation equipment to include:
   a. safe and proper stowage;
   b. periodic equipment inspections;
   c. cleaning of equipment;
   d. effective lubrication of equipment; and
   e. performing periodic maintenance.
4. Time: There are 8 x 50 minute periods, which are broken down as follows:
   a. 3 periods – navigation equipment;
   b. 1 period – EC 004;
   c. 1 period – post-EC review;
   d. 2 periods – PC 004; and

e. 1 period – post-PC review.

5. Method/Media:

a. lectures; and

b. practical exercises.

6. References:

A1, A2, A3, A12.

7. Training Aids:

a. equipment on the Navigation Section Supply Customer Account (SCA);

b. Navigation and Bridge Simulator (NABS);

c. classroom with PowerPoint;

d. TV/VCR;

e. cleaning material;

f. lubricating material; and

g. appropriate tools for conducting maintenance.

8. Test Details:

a. EC 002.03 named “Navigation Equipment - Practical Exercise”, is written in two 50 minute periods and consists of a practical/skill exercise; and

b. PC 002 assesses material covered in EO 002.03.

9. Remarks: None.

10. Teaching Points:

a. safely and correctly storing instruments ensuring they are:

(1) placed in correct cases; and

(2) secured for sea;

b. inspecting instruments for:

(1) cleanliness;

(2) damage; and

(3) requirements for repair.

c. cleaning instruments to include:

(1) lens and mirrors clear of obstructions;

(2) no salt build up; and

(3) free of rust or corrosion.

d. lubricating equipment to include:

(1) all parts move freely; and

(2) excess lubricant is removed.

e. performing periodic maintenance to ensure:

(1) all screws tightened;

(2) all straps secured;

(3) all lanyards are in proper working order;

(4) return of instruments requiring major repair or servicing to repair facility is initiated; and

(5) demands for replacement are initiated.

ENABLING CHECK - EC 002.03

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One EC booklet per trainee.

b. Classroom.

Arrangements And Procedures

1. This EC consists of written questions and a practical exercise to assess the trainee's knowledge of how to maintain navigation equipment.

2. References are not allowed.

3. Time allotted: 1 period (50 minutes).

4. Pre-Test Instructions for the CTO:

   a. trainees are to write their answers right on the EC;

   b. references are not allowed; and

   c. pass mark is 70%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of EC:

   a. knowledge;

   b. written;

   c. closed book; and

   d. short answer questions.

2. Time Limit: 1 period (50 minutes).

3. Pass Mark: 70%.

PART III - ENABLING CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
Identification of navigation equipment	25%
Maintaining navigation equipment	75%

PART IV – ENABLING CHECK LIST

Not Used.

PERFORMANCE CHECK - PC 002

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One PC per trainee.

b. Classroom.

Arrangements And Procedures

1. This written knowledge PC is used to assess the trainee's knowledge in logistical support of the Navigation section. It covers material from PO 002, 003 and 004.

2. Time: 2 periods (1 hour 40 minutes).

3. Pre-Exam Instructions for the CTO:

   a. trainees are to write their answers right on the PC;

   b. references are not allowed; and

   c. pass mark: 70%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of PC:

   a. knowledge;

   b. written;

   c. closed book; and

   d. short answer questions.

2. Material Covered: PO 002, PO 003 and PO 004.

3. Answers: All answers are to be written on the PC below the questions.

4. Time Limit: 2 periods (1 hour 40 minutes).

5. Pass Mark: 70%.

PART III - PERFORMANCE CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

CONTENT AREA	WEIGHT
EO 002.01 – Control Charts, Publications & Other	50%

CONTENT AREA	WEIGHT
Navigation Related Material	
EO 002.02 – Maintain the Navigation Section's Stores	25%
EO 002.03 – Maintain Navigation Equipment	25%

PART IV – PERFORMANCE CHECK LIST

Not Used.

PO 003 Execute the Navigating Officer's Requirements for Navigation Materials & Equipment.

EO 003.01 Maintain the OOW Notebook.

1. Performance:Maintain the OOW Notebook.

2. Conditions:

   a. given:

      (1) OOW notebook; and

      (2) HB pencils.

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to record information in the OOW notebook neatly and accurately to include:

   a. formatting the notebook;

   b. correcting mistakes;

   c. recording applicable entries as directed by the OOW or Navigator to include:

      (1) four-figure times;

      (2) courses;

      (3) speeds;

      (4) fixes;

      (5) ship's activities;

      (6) manoeuvring orders;

      (7) weather; and

      (8) any other miscellaneous entries.

   d. using proper terminology and authorized abbreviations; and

   e. recording and conducting time checks.

4. Time: There are 3 x 50 minute periods, which are broken down as follows:

   a. 1 period – Maritime Command policies on OOW notebooks; and

   b. 2 periods – practical exercise in NABS;

5. Method/Media:

   a. lecture; and

   b. practical exercise.

6. References:

A1, A2, A5, A6, A9, B2.

7. Training Aids:

   a. classroom with PowerPoint ;

   b. OOW notebooks and HB pencils;

   c. ship transit scenario for information to be recorded in the OOW notebook; and

   d. NABS.

8. Test Details: There is no EC for this EO. The material covered in this EO is assessed in PC 003.

9. Remarks:. None.

10. Teaching Points:

   a. explain why the OOW notebook is maintained to include:

      (1) its legal use; and

      (2) its use during investigations.

   b. the length of time notebooks are retained onboard; and

   c. when and where notebooks are sent for archiving;

   d. the Special Sea Duty Position of the Navigator's Yeoman

   e. formatting the OOW notebook;

   f. how to correct mistakes made in the notebook;

   g. information recorded in the OOW notebook to include:

      (1) closing up and securing of special parties;

      (2) changes in the ship's damage control condition state;

      (3) fixing the ship's position;

      (4) course changes;

      (5) speed changes;

(6) major ship events to include:

(a) flying stations;

(b) Replenishments at Sea (RAS);

(c) vertical replenishments at sea (VERTREPs);

(d) towing;

(e) weapons shoots;

(f) verification muster;

(g) alongsides; and

(h) departures.

(7) all emergencies;

(8) state of visibility;

(9) state of navigation lights;

(10) details of weather;

(11) details during berthing and unberthing;

(12) drawing an alongside diagram; and

(13) filling out the change of watch stamp.

EO 003.02 Maintain SDM Records.

1. Performance: Maintain SDM Records.

2. Conditions:

   a. given:

      (1) laptop with approved ECPINS software;; and

      (2) computer disk.

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to maintain SDM records to include:

   a. downloading recorded SDM voyage log data onto CD ROMs;

   b. storing the CD ROMs until the ship's logs have been sent to Director History and Heritage (DHH); and

   c. destroying the CD ROMs after the ship's logs have been sent to DHH.

4. Time: There is 3 x 50 minute period which is as follows:

   a. 1 period – SDM Record keeping; and

   b. 1 period – PC 002; and

   c. 1 period – post-PC review.

5. Method/Media: Lecture.

6. References:

A1, A2, A5, A6, A9, B2.

7. Training Aids:

   a. classroom with PowerPoint ;and

   b. SDM laptops.

8. Test Details: There is no EC for this EO. The material covered in this EO is assessed in PC 003.

9. Remarks: None.

10. Teaching Points:

    a. downloading recorded SDM voyage log data onto CD ROMs;

b. storing the CD ROMs on board until the ship's logs have been sent to DHH; and

c. when to destroy the CD ROMs.

PERFORMANCE CHECK – PC 003

PART I - ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One PC per trainee.

b. Classroom.

Arrangements And Procedures

1. This written knowledge and skill PC is used to assess the trainee's knowledge in maintenance of the OOW notebook and SDM Records. It covers material from EO 003.01 and EO 003.02.

2. Time: 1 period (50 minutes).

3. Pre-Exam Instructions for the CTO:

4. trainees are to write their answers on the PC;

5. references are not allowed;

6. This is a scenario based exam which will be presented by the CTO; and

7. pass mark: 80%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. Type of EC:

   a. knowledge;

   b. skill;

   c. written;

   d. closed book; and

   e. short answer questions.

2. Material Covered: EO 003.01 to include the trainee's understanding of what information is to be recorded in the OOW notebook, why it is recorded and how to record it. EO 003.02 to include the trainee's understanding of SDM record management procedures.

3. Answers: all answers to be written in the exam booklet.

4. Time Limit: 1 period (50 minutes).

5. Pass Mark: 80%

PART III - PERFORMANCE CHECK NOTES

1. The following table is to be used as a guideline for designating a value for each content area:

.CONTENT AREA	WEIGHT
OOW Notebook Theory	15%
OOW Notebook Practical (Skill)	70%
SDM Record Maintenance Procedures	15%

PART IV – PERFORMANCE CHECK LIST

Not Used.

EO 003.03 Set up and secure the bridge of HMC Ships with navigation required material and equipment

1. Performance: Set up and secure the bridge of HMC Ships with navigation required material and equipment.

2. Conditions:

   a. given:

      (1) a ship alongside with a complete navigation outfit; and

      (2) bridge set up/secure checklist.

   b. denied: assistance; and

   c. environmental: in all Conditions:

3. Standard: The trainee shall be able to:

   a. set-up the bridge of HMC Ships IAW ships' checklists to include:

      (1) ensuring all necessary equipment is in place and serviceable prior to sailing; and

      (2) ensuring all publication, charts, and stationary required for sailing are on the bridge;

   b. secure the bridge of HMC Ships in accordance with ships' checklists to include:

      (1) stowing all equipment, publications and stationary when Special Sea Dutymen (SSD) are secured and the vessel is alongside; and

      (2) scrubbing all charts and putting them back into their correct folios as directed by the Navigation Officer.

4. Time: There are 6 x 50 minute periods, which are broken down as follows:

   a. 1 period – setting up and securing the bridge;

   b. 2 periods – onboard ship alongside setting up and securing the bridge; and

   c. 3 periods – PCL 003.

5. Method/Media:

   a. lecture; and

   b. practical exercise.

6. References:

A1, A2, A11.

7. Training Aids:

   a. classroom with PowerPoint; and

   b. ships alongside with complete navigation outfits (preferably a HALIFAX class or KINGSTON class).

8. Test Details: There is no EC for this EO. The material covered in this EO is assessed in PCL 003.

9. Remarks: None.

10. Teaching Points:

   a. requirements for bridge set-up prior to sailing to include:

      (1) providing charts and publications as required/directed by the Navigating Officer;

      (2) supplying meteorological information as required;

      (3) providing required instruments, equipment and books for pilotage and navigation of the ship;

      (4) set up the bridge SDM IAW NOTM;

      (5) providing binoculars for the bridge team and lookouts; and

      (6) inspecting, lubricating and installing azimuth circles.

   b. requirements for securing the bridge once alongside to include:

      (1) stowing all binoculars, charts and navigation gear;

      (2) scrubbing all charts as required by the Navigating Officer;

      (3) shutting down SDM and completing the days records;

      (4) inspecting all navigation equipment and material to determine if anything requires maintenance or needs to be replaced.

PERFORMANCE CHECK LIST – PCL 003

PART I – ADMINISTRATIVE INSTRUCTIONS

Material And Facilities Required

a. One PCL per trainee.

b. HALIFAX class or KINGSTON class bridge.

Arrangements And Procedures

1. This practical skill PC is used to assess the trainee's ability to set up and secure the bridge.

2. Time: 3 periods (30 minutes per trainee).

3. Pre-Exam Instructions for the CTO:

   a. this PC is conducted using the Performance Checklist (PCL) located in Annex K;

   b. trainees must receive a checkmark beside all items in the PCL, which equates to a "Satisfactory" rating, in order to pass PC 006; and

   c. copies of the PCL should be issued to trainees at least one day prior to the PC.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. Type of PC:

   a. skill; and

   b. practical assessment using a Performance Checklist (PCL).

2. Material Covered: PO 006.

3. Time Limit: 30 minutes per trainee.

4. Pass Mark: checkmark beside each element of the PCL, which equates to an overall rating of "Satisfactory".

PART III – PERFORMANCE CHECK NOTES

1. See PCL 003 found in Annex K.

PART IV – PERFORMANCE CHECK LIST

Not Used.

ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication

A2 = Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

REF CODE	PUBLICATION	QTY
A1	Ships Standing Orders	available online
A2	MARCORDS	available online
A3	A-LM-185-002 / US-Z01 Static Unit Distribution Account Holder Supplementary Instructions	available online
A4	A-GN-H01-001/AG-001 Canadian Hydrographic Supplies Handbook	10
A5	Item 839 Section 1-3 Nautical Chart Symbols, Abbreviations And Related Matter	10
A6	CNM Canadian Navigation Manual	10
A7	Navigator's Yeoman Handbook	10
A8	DND & BA Weekly Corrections and Notice to Mariners	10 of each
A9	Ship's Log	1
A10	CIS/01/2 (D) Canadian Forces Instructions for COMSEC	available online
A11	C-57-007-002/AF –001 BR 45(1)	available online
A12	C-57-007-002/AF-002 BR 45(2)	available online

ANNEX B – TASK LIST

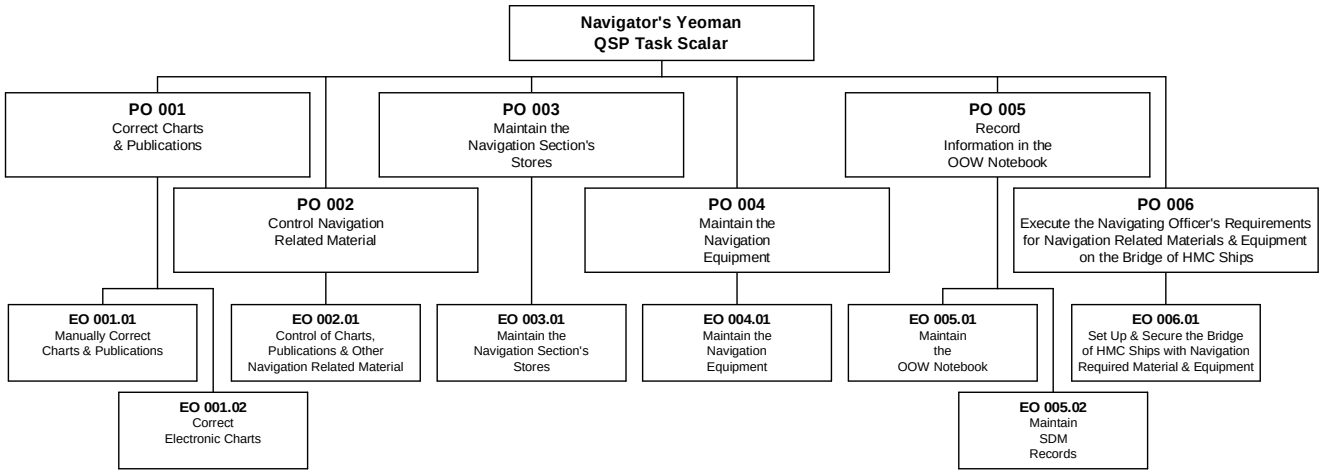
JBOS/OSS TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
1	Maintain navigation stores.	Trg	003
2	Control inventory of charts, publications and related material.	Trg	002
3	Correct charts/publications.	Trg	001
4	Record navigational information in the OOW notebook.	Trg	005
5	Order/return navigation equipment	Trg	004
6	Setup/secure bridge for navigation	Trg	006
7	Identify chart errors	Trg	001

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

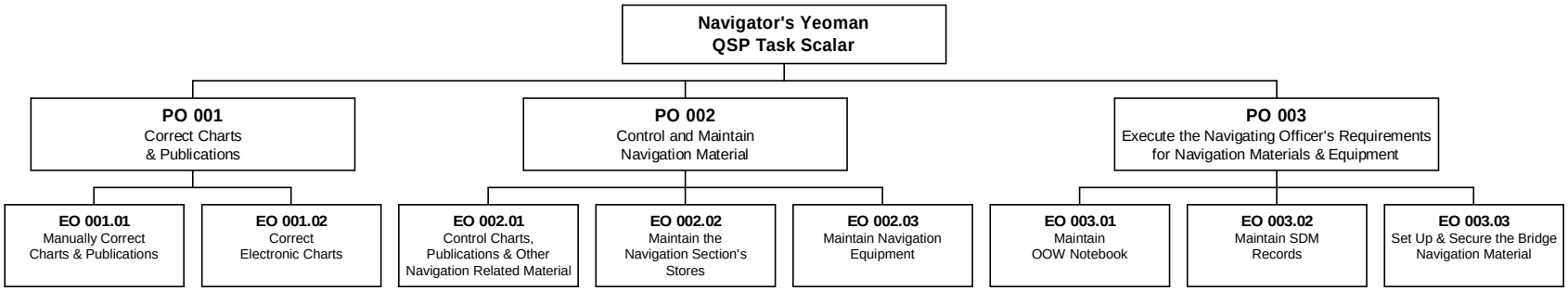
NOT USED

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR

Original QSP WB Scalar



Revised QSP Scalar



ANNEX E – ON-JOB PERFORMANCE RECORD

NOT USED

ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
A	Accountable Equipment
AGXO	Navigation Yeoman Course Code
ATT	Actual Training Time
AV	Available for Issue
BA	British Admiralty, abbreviation indicating charts, publications, etc., that originated in UK.
BAT	Base Administrative Time
C	Consumable
CANHYDROPAC/LANT	Canadian Hydrographic Pacific/Atlantic
CCA	Curriculum Control Authority
CF 377	Course Reports
CFAO	Canadian Forces Administrative Orders
CHN	Canadian Hydrographic Note
CHS	Canadian Hydrographic Service
CMS	Chief of Maritime Staff
CO	Commanding Officer
COX'N	Coxswain
CS	Contingency Stock
C Stds O	Chief Standards Officer
CTO	Course Training Officer
D	Disposable
DDNO	Deep Draught Navigating Officer
DGPS	Differential Global Positioning System
DHH	Director History and Heritage
DMTE	Director Maritime Training and Education

ACRONYM / ABBREVIATION	TERM/DEFINITION
DNC	Digital Nautical Chart
E	Controlled Equipment
EC	Enabling Check
ECDIS	Electronic Chart Display and Information System
ECL	Enabling Checklist
ECPINS (ECPINS-M)	Electronic Chart Precise Integrated Navigation System (Military)
ENC	Electronic Nautical Chart
EO	Enabling Objective
FNO	Fleet Navigating Officer
GPS	Global Positioning System
HSO	Hydrographic Services Office
HYDROPAC/LANT	Hydrographic Pacific/Atlantic
IAW	In Accordance With
IHO	International Hydrographic Organization
IMO	International Maritime Organization
INS	Inertial Navigation System
LORAN C	Long Range Radio Aid to Navigation System “C”
LPO	Local Purchase Order
MARCORD	Maritime Command Order
MAROPSGROUP	Maritime Operations Group
MARS	Maritime Surface and Subsurface
MCCR	Maritime Command Chart Register
MOS ID	Military Occupational Structure Identification
MSA	Material Service Account
MWVNO	Minor War Vessel Navigating Officer

ACRONYM / ABBREVIATION	TERM/DEFINITION
NABS	Navigation Bridge Simulator
NAVAREA	Navigation Area
Navigation Outfit	This is a series of charts and publications issued by Hydrographic Services Office (HSO).
NAVRESHQ	Naval Reserve Headquarters
NCIOP	Naval Combat Information Operator
NDHQ	National Defence Headquarters
NOTC	Naval Officers Training Centre
NTDSS	Naval Training Document Support System
O	Obsolete
ODAS	Ocean Data Acquisition System
OOW	Officer of the Watch
OSS AGJL	Occupation Speciality Specifications – Environmental Qualification - Sea/Surface (AGJL)
PC	Performance Check
PCL	Performance Checklist
PO	Performance Objective
QSP	Qualification Standard and Plan
QL#	Qualification Level #
RD	Restricted
RP	Repairable
RAS	Replenishment at Sea
S	Services
S 57	Standard 57 (vector format electronic chart)
SCA	Supply Customer Account – list of equipment and stores items for which a designated person is responsible.
SDM	SHINNADS Dual-M

ACRONYM / ABBREVIATION	TERM/DEFINITION
SHINNADS	Shipboard Integrated Navigation and Display System
SNI	Senior Navigation Instructor
SOP	Standard Operating Procedure
SSD	Special Sea Dutymen
StdsO Nav	Standards Officer – Navigation Representative
TAT	Training Administrative Time
TRB	Training Review Board
VERTREP	Vertical Replenishment at Sea
WGS 84	World Geodetic Survey 1984
XO	Executive Officer

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1 period = 50 min

Each training day is seven (7) periods from 0800-1550, this allows for 50 min in breaks and a one (1) hour lunch.

PO/EO	Number of Periods						
	Classroom Lecture	NABS	Visit Outside Agencies	NPTT (during class)	NPTT (after hours)	EC / PC Review (Pre/Post)	EC / PC
PO 001: Correct Charts & Publications							
EO 001.01 Manually Correct Charts & Publications	5		3				
EC 001.01 Manual Chart Correction Exercise						1	4 (classroom)
EO 001.02 Correct Electronic Charts	0			14			
EC 001.02 Electronic Chart correction Exercise						1	5 (classroom)
PC 001 Correct Charts and Publications						1	2 (classroom)
PO 001 TOTAL	5		3	14		3	11 (classroom)
PO 002: Control and Maintain Navigation Material							
EO 002.01 Control of Charts, Publications & Other Navigation Related Material	2						
EO 002.02 Maintain the Navigation Section's Stores	2						
EC 002.02 Navigation Section's Stores – Practical Exercise						1	1 (classroom)

EO 002.03 Maintain the Navigation Equipment	3						
EC 002.03 Navigation Equipment – Practical Exercise						1	1 (classroom)
PC 002 Control and Maintain Navigation Equipment						1	2 (classroom)
PO 002 TOTAL	7					3	4 (classroom)
PO 003: Execute the Navigating Officer's Requirements for Navigation Related Materials & Equipments							
EO 003.01 Maintain the OOW Notebook	1	2					
EO 003.02 Maintain SDM Records	1						
PC 003 Maintain Navigation and SDM Records						1	1 (classroom)
EO 003.03 Set Up & Secure the Bridge of HMC Ships with Navigation Required Material & Equipment	1		2				
PCL 003 Set Up & Secure Bridge							3 (ship alongside)
PO 003 TOTAL	3	2	2			1	4 (mixed)
TOTAL PERIODS	15	2	5	14	0	7	19 (mixed)

	Periods	Day(s)
Classroom = lectures + EC/PC Review + EC/PC	15+7+16=38	5.43
NABS = NABS + EC/PC	2+0=2	0.29
NPTT = NPTT + EC/PC	14+0=14	2
Outside Agencies = outside agencies + EC/PC	5+3=8	1.14
TAT	3	0.43

BAT	5	0.71
Stat Holiday	0	0.00
TOTAL TRAINING DAYS	70 periods	10 days

Nav Yeo CTO: Room: Week 1

	1	2	3	4	5	6	7
Trg Day	0800-0850	0900-0950	1000-1050	1100-1150	1300-1350	1400-1450	1500-1550
1	TAT CTO/ CO/XO/Stds/ COXN Intro	BAT In-Routine	BAT Stores Issue	001.01 Chart Symbols	002.02 CF Supply System (Guest Lecturer)		001.01 Geodesy
2	002 EC 002.02 Maintain Nav Stores	001.01 Geodesy	001.01 Manual Chart Corrections		001.01 Tour of HSO		
3	001.01 EC 001.01 Manual Chart Correction Exercise				003.01 Maritime Command Policies on OOW Notebook	003.01 NABS Practical Exercise	
4	003 PC 003 OOW Notebook	002.02 EC 003.01 Review	002.03 Navigation Equipment			002.01 Responsibilities of the Navigator's Yeoman	003.03 Set Up & Secure Bridge
5	002.03 EC 004.01 Nav Equipment Practical Exercise	001.01 EC 001.01 Review	003.03 Set Up & Secure Bridge		003 PCL 003 Set Up & Secure Bridge (HALIFAX or KINGSTON Class ship)		

Notes:

1.
2.
3.

Nav Yeo **CTO:** **Room:** **Week 2**

	1	2	3	4	5	6	7
Trg Day	0800-0850	0900-0950	1000-1050	1100-1150	1300-1350	1400-1450	1500-1550
6	001.02 SDM						
7	001.02 SDM						
8	002.01 Nav Admin & Inspections	003.02 Maintain SDM Records	001.02 EC 001.02 Electronic Chart Correction Exercise				
9	002 PC 002 Navigation Material, Stores & Equipment		TAT Sports		001 PC 001 Correct Charts & Publications		003 PC 003 Review
10	001 PC 001 Review	002 PC 002 Review	002.03 EC 002.03 Review	001.02 EC 001.02 Review	BAT Course Critique & Out Routine		

Notes:

1.
2.
3.

ANNEX H – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

EQUIPMENT AND FACILITIES			
ITEM	**BLDG NUMBER**	**TRAINING DAYS**	**REMARKS**
HALIFAX Class and/or KINGSTON Class vessel alongside.	N/A	½ day	Require access to the ship's bridge and navigation outfit to set up and secure the bridge with navigation related material and equipment.
NABS	WP 1094	½ day	Used to augment classroom instruction and conduct navigation passages.
Classroom	WP 1094	FC	Main Instruction area.
CBT Lab	WP 1094	A/R	N/A.
TV, VCR, Projector & Computer with Power Point	WP1094	FC	N/A.
Chart Tables	WP 1094	FC	One per trainee.
Chairs	WP 1094	FC	One per trainee.

Note: FC = Full Course A/R = As Required

STORES REQUIREMENTS		
ITEM DESCRIPTION	**QUANTITY (PER TRAINEE/COURSE)**	**NOTES**
Laptop	1 per trainee	With MS Office.
SDM Laptop	1 per trainee	
SDM Dongle	1 per trainee	
Plastic Parallel Ruler	1 per trainee	
Compass Set	1 per trainee	
Dividers Large	1 per trainee	
Dividers Small	1 per trainee	

Publications	As per Annex A.	Publications are outlined in Annex A.
CD-R (Box 10)	1 per course	
Plain Paper	1 box per course	
Binders 3 ring – 8 ½ x 11”	1 per trainee	
OOW Notebook	1 per trainee	
Pens	3 per trainee	
Pencils (2B leads)	1 box per trainee	
Erasers	3 per trainee	

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

DATE	COMMENT
15 MAY 05	DMTE approved chapter 1 and 2
07 JUN 05	VENTURE NOTC approved chapter 3 and chapter 4.
1 JAN 12	Editorial amendments have been made to reflect the new Maritime Training System construct and restoring the historic name of the Royal Canadian Navy.

ANNEX K – EVALUATION MATRIX

PCL 003 Set Up and Secure the Bridge Navigation Material (next page)

PCL 003 – Set Up & Secure the Bridge Navigation Material

Trainee:_____ Assessor:_____ Date:_____

Bridge Set Up Pre-Sail Checklist	**Score (Pass or Fail)**	**Comments/Observations**
1. On flash-up confirmed Met equipment operability with CSE department. (FFH only)		
2. Binoculars placed on the bridge. Ensured straps were checked and oculars moved freely.		
3. Gyro covers removed and Pelorus rings oiled, cleaned and set out.		
4. Setup the following equipment on the chart table:		
a. dividers;		
b. parallel;		
c. compass;		
d. STD ruler;		
e. OOW notebooks;		
f. OOW notebook stamp and stamp pad;		
g. erasers; and		
h. pencils.		
5. Stuart's distance meter placed on the bridge. Checked lanyard and ensured it was cleaned, oiled and setup properly.		
6. Hand held anemometer placed on the bridge. (MCDV only)		
7. Wind calculator on the bridge.		
8. Time check piped throughout the ship.		
9. Required Publications:		
a. CHP 3;		
b. CHP 5;		
c. CHP 7;		
d. Captains Standing Orders;		
e. Nautical Almanac;		
f. Norries Tables;		
g. Bridge Information Pack;		
h. Emergency Bridge Pack;		
i. List of Lights;		
j. Annual Notice to Mariners;		
k. Relevant Pilot / Sailing Directions;		
l. Tide Tables / Current Tables; and		
m. Collision Regulations.		
10. History tape recorder set. (MCDV only)		
11. Night vision goggles.		
12. SDM Set-up		

UNSAT / SAT

To achieve the requirements of PCL 006 the trainee must achieve a SAT standard. A SAT is awarded when the trainee has received a checkmark beside all items. Please circle the appropriate assessment above. A complete debrief has been conducted and the trainee has seen the content of this assessment form:

CTO's Signature: _____ Trainee's Signature: _____ Date:_____